Type 2: Application Project

Smart Home HAR for Speech-Impaired & Hearing-Impaired Users

23CSE399 – Project Phase 2
Guide Review 1 | Panel Review 1

Adhikkesh S K
Adithya Ajay
Amrith B
Gowreesh B
Rishiikesh S K

Guide: Ms. Nalinadevi K
Department of CSE, Amrita Vishwa Vidyapeetham

1. Use Cases: Recognizing Activities in a Privacy-Preserving Smart Home

Two real-world situations motivating WiFi CSI based HAR

UC1: Daily Activity Recognition

Scenario: Resident performs routine tasks (e.g., walking, sitting, cleaning).
Challenge: Cameras are intrusive; standard sensors lack fine-grained motion data.
WiFi CSI: Non-intrusively captures propagation changes caused by human movement.

UC2: Activity $\rightarrow$ Sign Transition

Scenario: User transitions from household activities to sign language.
Challenge: Gestures can easily be confused with ordinary daily movements.
WiFi CSI: Delivers fine-grained temporal data to reliably separate signs from daily tasks.

2. Use Cases: Multi-Channel CSI for Robust HAR

Extending CSI acquisition across non-overlapping WiFi channels

UC3: Multi-Channel Activity Sensing

Scenario: Monitoring activities across multiple non-overlapping WiFi channels.
Implementation: Sequential hopping (Channels 1, 6, 11) with 100 ms dwell time.
Purpose: Exploit frequency diversity to study and improve HAR robustness.
Limitation: Sequential channel hopping rather than simultaneous reception.

Key Insight:
Controlled SDR TX–RX allows the channel, packet generation, and sensing procedure to be custom-designed specifically for HAR.

3. Motivation: Why Move to a Controlled SDR-Based CSI System?

1. Privacy

Intrusive camera-based monitoring is avoided.

2. Fine-Grained Motion

Captures precise amplitude and phase shifts caused by human movement.

3. Controlled Acquisition

Avoids dependence on unpredictable phone or laptop network traffic.

4. Frequency Diversity

Enables sequential CSI collection across non-overlapping WiFi channels.

5. Reproducibility

Provides control over TX parameters, channel sequence, and dwell time.

Key Insight

SDR provides direct, programmatic control over physical-layer sensing.

4. Problem Statement & Research Objectives

Building a controlled, privacy-preserving CSI-HAR pipeline

Problem Statement:

Existing HAR systems rely on intrusive cameras, obtrusive wearables, or uncontrolled WiFi traffic. A controlled CSI framework is required to study privacy-preserving HAR and frequency-diverse WiFi sensing.

Research Objectives Completed:

- **Controlled CSI Acquisition:** Develop a dedicated SDR transmitter–receiver setup.
- **Channel Diversity:** Collect CSI while cycling through selected non-overlapping WiFi channels.
- **CSI Processing:** Extract, label, and store CSI corresponding to the active channel.
- **HAR:** Use temporal CSI variations for activity classification.

4. Problem Statement & Research Objectives

Objectives to be completed:

- **RO1:** To develop and evaluate a WiFi-based sensing approach for reliable fall detection.
- **RO2:** To collect and analyze multi-class human activity data and identify the most informative subcarriers for activity recognition.
- **RO3:** To investigate WiFi-based sensing for sign language identification and develop a suitable classification model.
- **RO4:** To explore a multimodal machine learning approach by integrating WiFi sensing with complementary sensing modalities for improved activity and fall detection.

5. Literature Survey

Paper (Year)	Method / Technique	Description
SignFi (2018)	9-layer CNN on WiFi CSI amplitude and phase	Recognizes 276 sign-language gestures with 98% lab accuracy; accuracy falls to 67% for users unseen during training.
Widar3.0 (2021)	Domain-independent Body-coordinate Velocity Profile (BVP) + CNN–GRU	Achieves zero-effort cross-domain gesture recognition (83–92%), but requires line-of-sight and multiple synchronized WiFi links.
ImgFi (2023)	CSI converted to Recurrence-Plot images, classified by a lightweight 2D-CNN	Reaches 99.5–99.8% accuracy with $\sim 90\times$ fewer parameters than ResNet50, but is evaluated within-domain only.
Joint Activity & Localization (2019)	Dual-branch 1D-CNN (ResNet1D) on CSI subcarrier fingerprints	Jointly predicts activity (88%) and indoor location (95.7%) from CSI, but on a single-user, single-room dataset.

Takeaway: Existing WiFi CSI systems are single-purpose (gesture-only or localization-only) and environment-specific. Our controlled multi-channel CSI pipeline aims to unify activity and sign-transition recognition in one privacy-preserving system.

6. System Evolution: From Passive Capture to Controlled CSI

The major implementation change in the current phase

SETUP 1

Phone Hotspot

↓

Laptop WiFi Traffic

↓

Single USRP
Passive CSI Capture

- Passive observation
- Traffic dependent
  - Limited channel control



SETUP 2

USRP N210 TX

↓

Wireless Channel

↓

USRP N210 RX

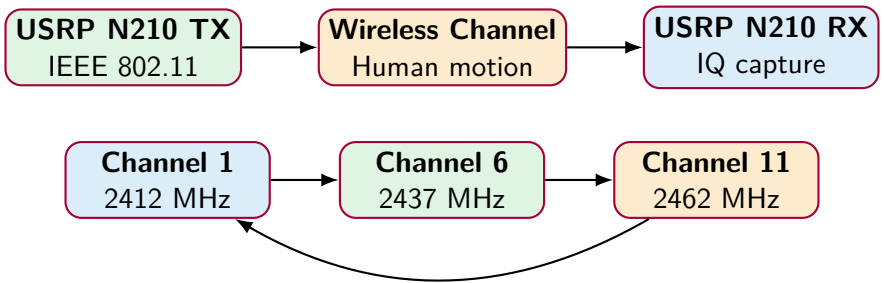
↓

CSI Extraction

- Controlled transmission
- Dedicated TX/RX
  - Channel hopping

7. Current Implementation: Two-USRP CSI Setup

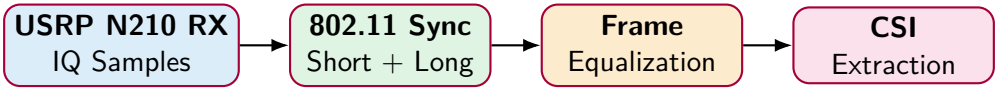
Controlled IEEE 802.11 transmission and sequential channel hopping



- Controlled IEEE 802.11 transmission
- Dedicated SDR transmitter and receiver
- Sequential hopping across Channels 1, 6 and 11
- ≈100 ms dwell per channel

8. CSI Acquisition Pipeline

Processing received 802.11 frames to extract CSI



RX Processing

802.11 synchronization
FFT and frame equalization
Custom CSI extraction

CSI Output

Complex channel response
Across OFDM subcarriers
$\rightarrow$ CSI samples for HAR

Key result: Received WiFi frames are converted into CSI measurements for subsequent HAR analysis.

9. Current Capability vs. Future Extension

Clarifying what the present implementation achieves

CURRENT

✓ Two-USRP TX–RX architecture
✓ Controlled WiFi transmission
✓ CSI extraction
✓ Channels 1, 6, 11
✓ Sequential hopping
✓ Channel-labelled CSI

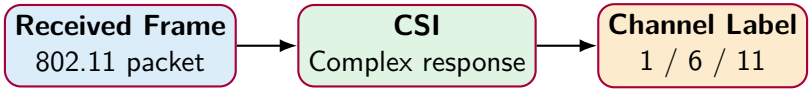
FUTURE

→ Sign-language gesture detection
→ Wearable-based identity tagging
→ WiFi CSI + wearable fusion
→ Multi-modal HAR
→ End-to-end activity/sign recognition

Important: The present two-USRP system hops between channels; it does **not** receive Channels 1, 6 and 11 simultaneously.

10. CSI Data: From Channel Response to HAR Input

What is extracted from each received WiFi frame



CSI Representation

Complex channel response measured across the OFDM subcarriers.

$$H_k = |H_k| e^{j\angle H_k}$$

Amplitude and phase provide the sensing information.

HAR Input

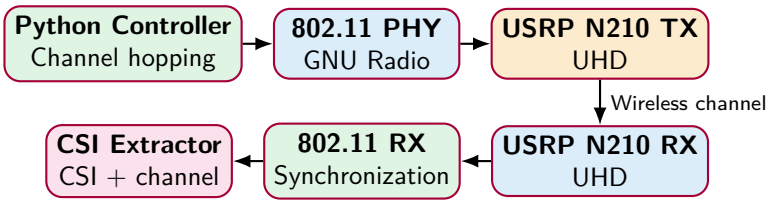
CSI samples are arranged over time to capture motion-induced channel variations.

Input: temporal CSI variations
Output target: activity / sign class

Key result: Each CSI sample is associated with the active WiFi channel, enabling

11. CSI Processing and Software Architecture

GNU Radio + UHD + Python channel control + custom CSI extraction



TX / Channel Control

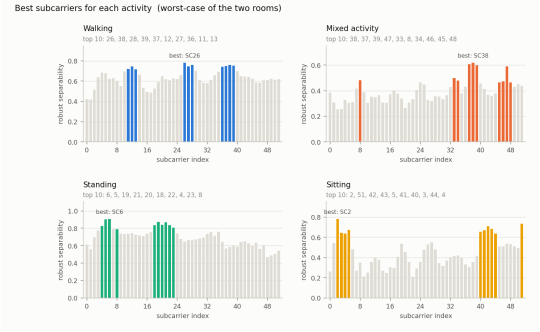
- ✓ Python-controlled frequency hopping
- ✓ GNU Radio 802.11 PHY
- ✓ UHD control of USRP N210 TX
- ✓ Channels 1, 6 and 11

RX / CSI Processing

- ✓ USRP N210 IQ reception
- ✓ 802.11 synchronization
- ✓ Frame equalization
- ✓ Custom CSI extraction

Current output: CSI measurements associated with the active channel.

12. Optimal Subcarrier Selection per Activity



Caption: Subcarriers were ranked by how much they respond to motion versus noise, kept only if that held true across both rooms and all subjects, thinned to remove duplicates, and confirmed by testing on a person the model hadn't seen.

12. Active vs. Passive CSI Sensing

Two paradigms for acquiring WiFi Channel State Information

Passive Sensing



How it works:

- Phone and laptop are the real TX/RX pair
- USRP transmits nothing; it only listens
- CSI is overheard from existing traffic
- Traffic is uncontrolled, app-dependent

Active Sensing



How it works:

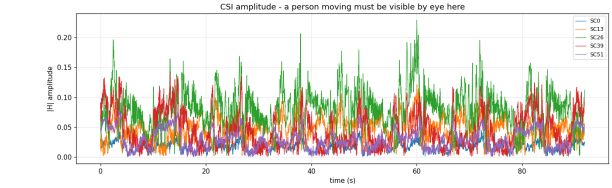
- One USRP actively transmits 802.11 frames
- The second USRP passively receives & extracts CSI
- Channel, rate, and timing are fully controlled

13. How the Data Was Collected: Walking & Standing

Left: measured CSI amplitude for that take. Right: the activity being performed.

Walking

action take · Amrith · Room A





Standing

no-action take · Rishiikesh · Room A



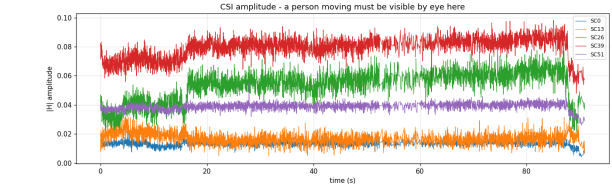


14. How the Data Was Collected: Sitting & Mixed Activity

Left: measured CSI amplitude for that take. Right: the activity being performed.

Sitting

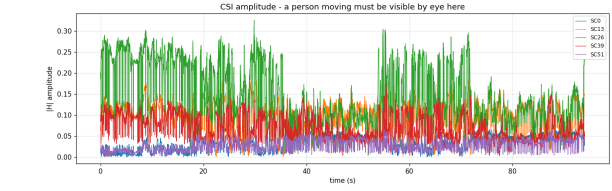
no-action take · Rishiikesh · Room A

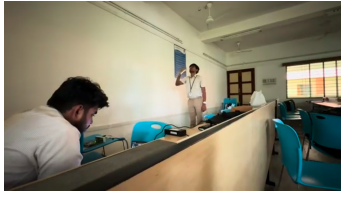




Mixed activity

action take · Gowreesh · Room B





15. Limitations and Future Work

What remains before full sign-language recognition

Current Limitations

Channel Hopping
Sequential rather than simultaneous acquisition.

Synchronization
Independent SDRs require stable timing/frequency.

Dataset
Large labelled HAR/sign dataset is still required.

Next Phase

→ CSI pre-processing + features
→ HAR / sign recognition
→ WiFi CSI + wearable fusion

Long-term goal: Sign-language recognition with identity-aware multi-modal sensing.

16. References

1. D. J. Cook et al., *CASAS: A Smart Home in a Box*, Computer, 2013.
2. H. Alemdar et al., *ARAS human activity datasets in multiple homes*, PervasiveHealth, 2013.
3. D. Anguita et al., *A Public Domain Dataset for Human Activity Recognition*, ESANN, 2013.
4. R. Chavarriaga et al., *The Opportunity challenge*, Pattern Recognition Letters, 2013.
5. D. Micucci et al., *UniMiB SHAR*, Applied Sciences, 2017.
6. Y. Zheng et al., *Widar3.0: Zero-Effort Cross-Domain Gesture Recognition*, IEEE TPAMI, 2021.
7. G. Zhu et al., *CSI-Bench*, arXiv, 2025.
8. S. Huang et al., *WiMANS*, arXiv, 2024.
9. Y. Ma et al., *SignFi: Sign Language Recognition Using WiFi*, ACM IMWUT, 2018.
10. J. Li et al., *WiFinger: Finger Gesture Recognition Using CSI*, UbiComp, 2016.
11. B. D. Minor et al., *Learning Activity Predictors from Sensor Data*, IEEE TKDE, 2017.

Implementation basis

GNU Radio, UHD, IEEE 802.11 PHY/MAC processing, and custom CSI extraction.

Thank You

Smart Home HAR for
Speech-Impaired & Hearing-Impaired Users

Guide Review 1 | Panel Review 1

Department of Computer Science and Engineering
Amrita Vishwa Vidyapeetham